

Title: SiteBuilder AI- Enhanced Comprehensive Website Development Framework

Directory Structure for

SiteBuilder AI- Enhanced Comprehensive Website Development Framework\

```
|— .vscode/
|   |— index1.html
|   |— launch.json
|   └— settings.json
|— database.db
|— main.py
|— static/
|   |— AI.css
|   |— domain.css
|   |— homepage.css
|   |— landing.css
|   |— login.css
|   |— que1.css
|   |— styles.css
|   |— temp.css
|   |— Images/
|       |— drag.jpg
|       └— register.jpg
|— template/
|   |— AI.html
|   |— domain.html
|   |— homepage.html
|   |— landing.html
|   |— login.html
|   └— que1.html
```

- | | └─ register.html
- | | └─ temp.html
- | └─ trail2/
- | | └─ .vscode/
- | | | └─ index1 (2).html
- | | | └─ index1.html
- | | | └─ launch (2).json
- | | | └─ launch.json
- | | └─ backend/
- | | | └─ package.json
- | | └─ index.html
- | | └─ main.js
- | | └─ node_modules/ (Large number of dependencies)
- | | └─ website-builder/
- | | | └─ public/
- | | | └─ src/
- | | | | └─ components/
- | | | | └─ pages/
- | | | | └─ styles/
- | | | | └─ App.js
- | | | | └─ index.js
- | | | └─ package.json
- | | | └─ README.md

main.py

<!DOCTYPE html>

<html lang="en">

```
<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Select Domain</title>

  <link rel="stylesheet" href="../static/domain.css"> <!-- CSS relative path -->

</head>

<body>

  <div class="container">

    <h1>Select Domain</h1>

    <!-- The form redirects to index.html in the trail2 folder -->

    <form action="../trail2/index.html" method="get">

      <label class="radio-option">

        <input type="radio" name="domain" value="education" required>

        Education

      </label>

      <label class="radio-option">

        <input type="radio" name="domain" value="health" required>

        Health

      </label>

      <label class="radio-option">

        <input type="radio" name="domain" value="medical" required>

        Medical

      </label>

      <label class="radio-option">

        <input type="radio" name="domain" value="ecommerce" required>

        E-Commerce

    </form>

  </div>

</body>

</html>
```

```
        </label>

        <button type="submit">Next</button> <!-- Navigates to index.html -->

    </form>

</div>

</body>

</html>
```

main.js

```
const editor = grapesjs.init({
  container: '#editor',
  height: '100vh',
  width: 'auto',
  fromElement: true,
  storageManager: false,

  blockManager: {
    appendTo: '#blocks',
    blocks: [
      // Section Block
      {
        id: 'section',
        label: '<b>Section</b>',
        attributes: { class: 'gjs-block-section' },
        content: `<section>
          <h1>Section Title</h1>
          <p>This is a section block</p>
```

```
        </section>`,
    category: 'Basic',
    draggable: true,
    droppable: true,
  },
  // Text Block
  {
    id: 'text',
    label: 'Text',
    content: '<div data-gjs-type="text">Insert your text here</div>',
    category: 'Basic',
    draggable: true,
    droppable: true,
  },
  // Image Block
  {
    id: 'image',
    label: 'Image',
    content: { type: 'image', src: 'https://via.placeholder.com/150' },
    category: 'Media',
    draggable: true,
    droppable: false,
  },
  // Video Block
  {
    id: 'video',
```

```
    label: 'Video',
    content: {
      type: 'video',
      src: 'https://www.w3schools.com/html/mov_bbb.mp4',
    },
    category: 'Media',
    draggable: true,
    droppable: false,
  },
  // Button Block
  {
    id: 'button',
    label: 'Button',
    content: '<button class="gjs-btn">Click me</button>',
    category: 'Basic',
    draggable: true,
    droppable: false,
  },
  // Link Block
  {
    id: 'link',
    label: 'Link',
    content: '<a href="#">This is a link</a>',
    category: 'Basic',
    draggable: true,
    droppable: true,
```

```
},  
// List Block  
{  
  id: 'list',  
  label: 'List',  
  content: `

  
    <li>Item 1</li>  
    <li>Item 2</li>  
    <li>Item 3</li>  
  </ul>`,  
  category: 'Basic',  
  draggable: true,  
  droppable: true,  
},  
// Form Block  
{  
  id: 'form',  
  label: 'Form',  
  content: `  
    <label for="name">Name:</label>  
    <input type="text" id="name" name="name"><br><br>  
    <label for="email">Email:</label>  
    <input type="email" id="email" name="email"><br><br>  
    <button type="submit">Submit</button>  
  </form>`,  
  category: 'Forms',
```

```
    draggable: true,
    droppable: true,
  },
  // Input Field Block
  {
    id: 'input',
    label: 'Input Field',
    content: '<input type="text" placeholder="Enter text"/>',
    category: 'Forms',
    draggable: true,
    droppable: true,
  },
  // Textarea Block
  {
    id: 'textarea',
    label: 'Textarea',
    content: '<textarea placeholder="Enter text"></textarea>',
    category: 'Forms',
    draggable: true,
    droppable: true,
  },
  // Select Box Block
  {
    id: 'select',
    label: 'Select Box',
    content: `<select>
```



```
        <option value="1">Option 1</option>
        <option value="2">Option 2</option>
    </select>`,
    category: 'Forms',
    draggable: true,
    droppable: true,
},
// Checkbox Block
{
    id: 'checkbox',
    label: 'Checkbox',
    content: '<input type="checkbox"/> Checkbox',
    category: 'Forms',
    draggable: true,
    droppable: true,
},
// Radio Button Block
{
    id: 'radio',
    label: 'Radio Button',
    content: '<input type="radio" name="option"/> Radio Button',
    category: 'Forms',
    draggable: true,
    droppable: true,
},
// Heading Block
```

```
{
  id: 'heading',
  label: 'Heading',
  content: '<h2>Heading Title</h2>',
  category: 'Basic',
  draggable: true,
  droppable: true,
},
// Quote Block
{
  id: 'quote',
  label: 'Quote',
  content: '<blockquote>This is a quote</blockquote>',
  category: 'Typography',
  draggable: true,
  droppable: true,
},
// Icon Block
{
  id: 'icon',
  label: 'Icon',
  content: '<i class="fa fa-star" aria-hidden="true"></i>',
  category: 'Basic',
  draggable: true,
  droppable: false,
},
```

```
// Row Block
{
  id: 'row',
  label: 'Row',
  content: '<div class="row">Row content here</div>',
  category: 'Layout',
  draggable: true,
  droppable: true,
},
],
},
});
```

```
// Preview button functionality
```

```
document.getElementById('preview').addEventListener('click', () => {
  const win = window.open('', '_blank');
  win.document.write(editor.getHtml());
  win.document.write('<style>' + editor.getCss() + '</style>');
});
```

```
// View code button functionality
```

```
document.getElementById('viewCode').addEventListener('click', () => {
  alert(editor.getHtml() + '\n\n' + editor.getCss());
});
```

```
// Clear button functionality
```

```
document.getElementById('clear').addEventListener('click', () => {  
    editor.setComponents("");  
});
```

index.html

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
    <meta charset="UTF-8">  
    <meta http-equiv="X-UA-Compatible" content="IE=edge">  
    <meta name="viewport" content="width=device-width, initial-scale=1.0">  
    <title>GrapesJS Drag and Drop Builder</title>  
  
    <!-- External CSS (like Font Awesome for icons) -->  
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.7.0/css/font-awesome.min.css">  
    <link rel="stylesheet" href="style.css">  
  
    <!-- GrapesJS CSS -->  
    <link rel="stylesheet"  
href="https://cdnjs.cloudflare.com/ajax/libs/grapesjs/0.17.26/css/grapes.min.css">  
</head>  
<body>  
  
    <!-- Top Panel for Controls (Centered) -->  
    <div class="panel__top">
```

```
<button id="preview">Preview</button>
<button id="viewCode">View Code</button>
<button id="clear">Clear</button>
</div>

<!-- Container for the layout -->
<div class="container">
  <!-- Blocks Section (Left Panel) -->
  <div id="blocks">
    <!-- New block with image -->
    <div class="gjs-block">
      
      <p>Drag Image Block</p>
    </div>
  </div>

  <!-- Editor Container (Main Canvas Area) -->
  <div id="editor"></div>
</div>

<!-- GrapesJS and JavaScript -->
<script
src="https://cdnjs.cloudflare.com/ajax/libs/grapesjs/0.17.26/grapes.min.js"></
script>

<script src="main.js"></script>
```

```
</body>
```

```
</html>
```

style.css

```
/* General reset and layout */
```

```
body, html {
```

```
    margin: 0;
```

```
    padding: 0;
```

```
    height: 100%;
```

```
    display: flex;
```

```
    flex-direction: column; /* Stack elements vertically */
```

```
}
```

```
/* Container for blocks and editor */
```

```
.container {
```

```
    display: flex; /* Align blocks and editor side by side */
```

```
    flex-grow: 1; /* Take the remaining height */
```

```
}
```

```
/* Blocks section */
```

```
#blocks {
```

```
    width: 250px;
```

```
    padding: 10px;
```

```
    background-color: #f5f5f5;
```

```
    border-right: 1px solid #ddd;
```

```
    overflow-y: auto;
```

```
}
```

```
/* Editor area */
```

```
#editor {
```

```
    flex-grow: 1;
```

```
    height: 100vh; /* Full height of the viewport */
```

```
    border: 1px solid #ddd;
```

```
    background: #fff;
```

```
}
```

```
/* Block appearance */
```

```
.gjs-block {
```

```
    cursor: pointer;
```

```
    border: 1px solid #ddd;
```

```
    padding: 10px;
```

```
    background-color: #fff;
```

```
    transition: background-color 0.3s ease, box-shadow 0.3s ease;
```

```
}
```

```
.gjs-block:hover {
```

```
    background-color: #f0f0f0;
```

```
    box-shadow: 0 2px 6px rgba(0, 0, 0, 0.2);
```

```
}
```

```
/* Style for blocks inside the canvas */
```

```
.gjs-cv-canvas {
```

```
background: #f9f9f9;
}
```

```
/* Control panel styles */
```

```
.panel__top {
  padding: 10px;
  text-align: center;
  background-color: #f5f5f5;
  border-bottom: 1px solid #ddd;
}
```

```
.panel__top {
  display: flex;
  justify-content: center; /* Center the buttons */
}
```

```
.panel__top button {
  margin: 0 10px;
  padding: 8px 15px;
  cursor: pointer;
  background-color: #007bff;
  color: #fff;
  border: none;
  border-radius: 5px;
  font-size: 14px;
  transition: background-color 0.3s ease;
```



```
}
```

```
.panel__top button:hover {  
    background-color: #0056b3;  
}
```

```
/* Accessibility focus style */  
.panel__top button:focus {  
    outline: 2px solid #0056b3;  
}
```

```
/* Responsive adjustments */  
@media (max-width: 768px) {  
    #blocks {  
        width: 100%;  
        height: 150px;  
        border-right: none;  
        border-bottom: 1px solid #ddd;  
    }  
}
```

```
#editor {  
    height: calc(100vh - 150px);  
}
```

```
.panel__top {  
    top: 150px;
```

```
left: 0;  
right: 0;  
}
```

About project:

SiteBuilder AI is an advanced, AI-powered website development framework that simplifies the process of website creation for users without coding expertise. The platform integrates an intuitive drag-and-drop interface, AI-driven design assistance, and extensive customization options to enable users to build fully functional, visually appealing websites efficiently. The system offers a variety of pre-designed templates, real-time content management, and optimization tools for SEO and social media integration, making web development accessible to individuals, businesses, and organizations.

Workflow:

1. User Registration & Login: Users create an account or log in to access the SiteBuilder AI platform. A questionnaire is presented to understand user preferences.
2. Template Selection: Users can select from a range of pre-designed templates tailored to different industries.
3. Drag-and-Drop Editor: The interface allows users to add, move, and customize elements like text, images, buttons, and forms without any coding.
4. AI-Assisted Design Optimization: The AI suggests design improvements, layout arrangements, and content optimizations based on user input.

5. **Real-Time Preview & Editing:** Users can preview their website in real time, make modifications, and fine-tune the design.
6. **Code Generation:** The system automatically generates clean, optimized HTML, CSS, and JavaScript code. Users can also view and edit the code if needed.
7. **SEO & Performance Enhancement:** Built-in SEO tools help optimize metadata, keywords, and image alt texts, improving search engine visibility.
8. **Website Deployment:** Once satisfied, users can publish their website on a custom domain with integrated hosting options.
9. **Analytics & Monitoring:** Users can track visitor engagement, traffic sources, and performance metrics through an integrated dashboard.

Real-World Applications:

1. **Small Businesses:** SiteBuilder AI allows small business owners to create and maintain an online presence without hiring professional developers.
2. **E-Commerce:** The platform includes e-commerce-friendly templates, enabling users to build online stores with seamless payment gateway integration.
3. **Marketing & Advertising:** Marketers can design landing pages and promotional sites with AI-optimized layouts for higher conversion rates.
4. **Portfolio Websites:** Freelancers and artists can showcase their work using customizable templates that highlight their skills and creativity.

5. **Educational Platforms:** Educators can create e-learning websites, online course platforms, and school portals with interactive elements.
6. **Corporate Websites:** Companies can build professional-grade websites with AI-driven content management and analytics tracking.
7. **Non-Profit Organizations:** Charities and NGOs can create awareness websites, accept donations, and share their missions effectively.

Summary:

SiteBuilder AI revolutionizes website development by providing an intuitive, AI-powered platform for users of all skill levels. By integrating a drag-and-drop interface, AI-assisted design recommendations, and robust customization features, it eliminates the barriers of traditional web development. With applications spanning various industries, from business and education to e-commerce and marketing, SiteBuilder AI ensures accessibility, efficiency, and scalability in modern website creation. Future improvements could include enhanced AI capabilities, multilingual support, and expanded e-commerce functionalities to further empower users in the digital landscape.
